

Tic-Tac-Toe Game

Project Title:

Tic-Tac-Toe: A Responsive Web-based Game using HTML, CSS & JavaScript

Objective:

The objective of this project is to design and develop an interactive, mobile-responsive Tic-Tac-Toe game using modern web technologies. The game allows two players to compete by placing their respective marks (X and O) on a 3×3 grid. The first player to align three marks in a row, column, or diagonal wins the game.

Tools & Technologies Used:

Technology Purpose

HTML5	Structure and layout of the game board and interface
CSS3	Styling, responsive layout, background gradient, and UI enhancements
JavaScript	Game logic, win/draw detection, turn handling, and pop-up interactions

User Interface Overview:

- The background uses a vibrant **linear gradient** to enhance visual appeal.
- The title "**Tic-Tac-Toe**" is prominently displayed at the top in a stylish and bold font.
- A **3×3 board** is placed at the center, with clearly styled clickable boxes (cells).
- Below the board, a **status text** dynamically shows which player's turn it is.

- A **pop-up window** appears at the end of each game, announcing the result:
 - 🏆 Trophy icon for the winning player
 - 🤝 Handshake icon for a draw
 - Two buttons are available:
 - **Restart Game:** Clears the board and continues with the same session.
 - **New Game:** Reloads the game from the beginning.
-

⚙️ Game Functionality:

1. Turn Management:

- The game starts with Player X.
- Players alternate turns by clicking empty cells on the board.

2. Move Validation:

- A player cannot select a cell that is already filled.

3. Win Detection:

- After each move, the game checks if the current player has won.
- A win is detected if the same mark appears in a full row, column, or diagonal.

4. Draw Detection:

- If all cells are filled and no player has won, a draw is declared.

5. Pop-Up Message:

- If there's a winner, a **trophy icon** and the winner's name are displayed.
- In case of a draw, a friendly message with a **handshake icon** is shown.

6. Game Reset Options:

- **Restart Game:** Clears the board for another round with the same players.
- **New Game:** Reloads the page to start from scratch.

✅ Key Learning Outcomes:

- Gained hands-on experience with **DOM manipulation** using JavaScript.
- Applied CSS concepts to create a **modern and visually appealing UI**.
- Understood the use of **event listeners, conditionals, and arrays** in building game logic.
- Learned how to implement user feedback through **pop-ups and dynamic text**.
- Applied **responsive design principles** to make the game functional across devices.

Code:

HTML CODE

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Tic-Tac-Toe</title>
  <link rel="stylesheet" href="style.css"> <!-- Link to CSS File -->
</head>
<body>
  <div class="game-container">
    <h1 class="game-title">Tic-Tac-Toe</h1>
```

```
<!-- Game Board -->
```

```
<div class="board">
```

```
  <div class="cell" data-index="0"></div>
```

```
  <div class="cell" data-index="1"></div>
```

```
  <div class="cell" data-index="2"></div>
```

```
  <div class="cell" data-index="3"></div>
```

```
  <div class="cell" data-index="4"></div>
```

```
  <div class="cell" data-index="5"></div>
```

```
  <div class="cell" data-index="6"></div>
```

```
  <div class="cell" data-index="7"></div>
```

```
  <div class="cell" data-index="8"></div>
```

```
</div>
```

```
<!-- Player Turn Status -->
```

```
<div id="status">Player X's Turn</div>
```

```
<!-- Restart Button -->
```

```
<button onclick="restartGame()">Restart Game</button>
```

```
</div>
```

```
<!-- Winner Popup -->
```

```
<div id="popup" class="popup">
```

```
  <p id="winner-text"></p>
```

```
  <button onclick="closePopup()">New Game</button>
```

```
</div>
```

```
<script src="script.js"></script> <!-- Link to JavaScript File -->
</body>
</html>
```

CSS CODE:

```
/* Background - Gradient */
body {
  font-family: 'Arial', sans-serif;
  text-align: center;
  background: linear-gradient(to right, #8e2de2, #4a00e0);
  height: 100vh;
  display: flex;
  flex-direction: column;
  align-items: center;
  justify-content: center;
  margin: 0;
  color: white;
}

/* Game Container */
.game-container {
  display: flex;
  flex-direction: column;
  align-items: center;
  justify-content: center;
}
```

```
/* Game Title at the Top */
```

```
.game-title {  
    font-size: 50px;  
    font-weight: bold;  
    text-transform: uppercase;  
    letter-spacing: 3px;  
    margin-bottom: 20px;  
    text-shadow: 2px 2px 10px rgba(0, 0, 0, 0.5);  
}
```

```
/* Tic-Tac-Toe Board */
```

```
.board {  
    display: grid;  
    grid-template-columns: repeat(3, 100px);  
    gap: 10px;  
    background: rgba(255, 255, 255, 0.1);  
    padding: 15px;  
    border-radius: 15px;  
}
```

```
/* Individual Cells */
```

```
.cell {  
    width: 100px;  
    height: 100px;  
    background: white;
```

```
color: #4a00e0;
font-size: 32px;
font-weight: bold;
border-radius: 10px;
display: flex;
align-items: center;
justify-content: center;
box-shadow: 2px 2px 10px rgba(0, 0, 0, 0.2);
cursor: pointer;
transition: 0.3s;
}
```

```
.cell:hover {
  background: #eee;
}
```

```
/* Player Turn Status at Bottom */
#status {
  font-size: 18px;
  font-weight: bold;
  background: rgba(255, 255, 255, 0.2);
  padding: 10px 20px;
  border-radius: 10px;
  display: inline-block;
  margin-top: 20px;
}
```

```
/* Buttons */
```

```
button {  
    padding: 12px 24px;  
    font-size: 16px;  
    font-weight: bold;  
    margin-top: 20px;  
    background: black;  
    color: white;  
    border: none;  
    cursor: pointer;  
    border-radius: 8px;  
    transition: 0.3s;  
}
```

```
button:hover {  
    background: #222;  
}
```

```
/* Popup Winner Message */
```

```
.popup {  
    display: none;  
    position: fixed;  
    top: 50%;  
    left: 50%;  
    transform: translate(-50%, -50%);
```



```
background: white;
color: black;
padding: 20px;
border-radius: 10px;
box-shadow: 0px 4px 10px rgba(0, 0, 0, 0.3);
text-align: center;
font-size: 22px;
}
```

```
/* Popup Button */
.popup button {
margin-top: 10px;
background: #4a00e0;
color: white;
}
```

JAVASCRIPT CODE:

```
let currentPlayer = "X";
let board = ["", "", "", "", "", "", "", "", ""];
let gameActive = true;
const cells = document.querySelectorAll(".cell");
const statusText = document.getElementById("status");
const popup = document.getElementById("popup");
const winnerText = document.getElementById("winner-text");
const winnerIcon = document.getElementById("winner-icon"); // Added icon
element
```

```

const winningCombos = [
  [0, 1, 2], [3, 4, 5], [6, 7, 8],
  [0, 3, 6], [1, 4, 7], [2, 5, 8],
  [0, 4, 8], [2, 4, 6]
];

// Handle Cell Click
cells.forEach(cell => {
  cell.addEventListener("click", () => {
    const index = cell.getAttribute("data-index");
    if (board[index] === "" && gameActive) {
      board[index] = currentPlayer;
      cell.textContent = currentPlayer;
      checkWinner();
      currentPlayer = currentPlayer === "X" ? "O" : "X";
      statusText.textContent = `Player ${currentPlayer}'s Turn`;
    }
  });
});

// Check for Winner
function checkWinner() {
  for (let combo of winningCombos) {
    let [a, b, c] = combo;
    if (board[a] && board[a] === board[b] && board[a] === board[c]) {
      gameActive = false;
    }
  }
}

```

```

        showPopup(`Player ${board[a]} Wins! 🏆`);
        return;
    }
}

if (!board.includes("")) {
    gameActive = false;
    showPopup("It's a Draw! 🤝");
}
}

```

// Show Winner Popup with Icon

```

function showPopup(message) {
    winnerText.innerHTML = message;
    popup.style.display = "block";
}

```

// Close Popup and Restart Game

```

function closePopup() {
    popup.style.display = "none";
    restartGame();
}

```

// Restart Game

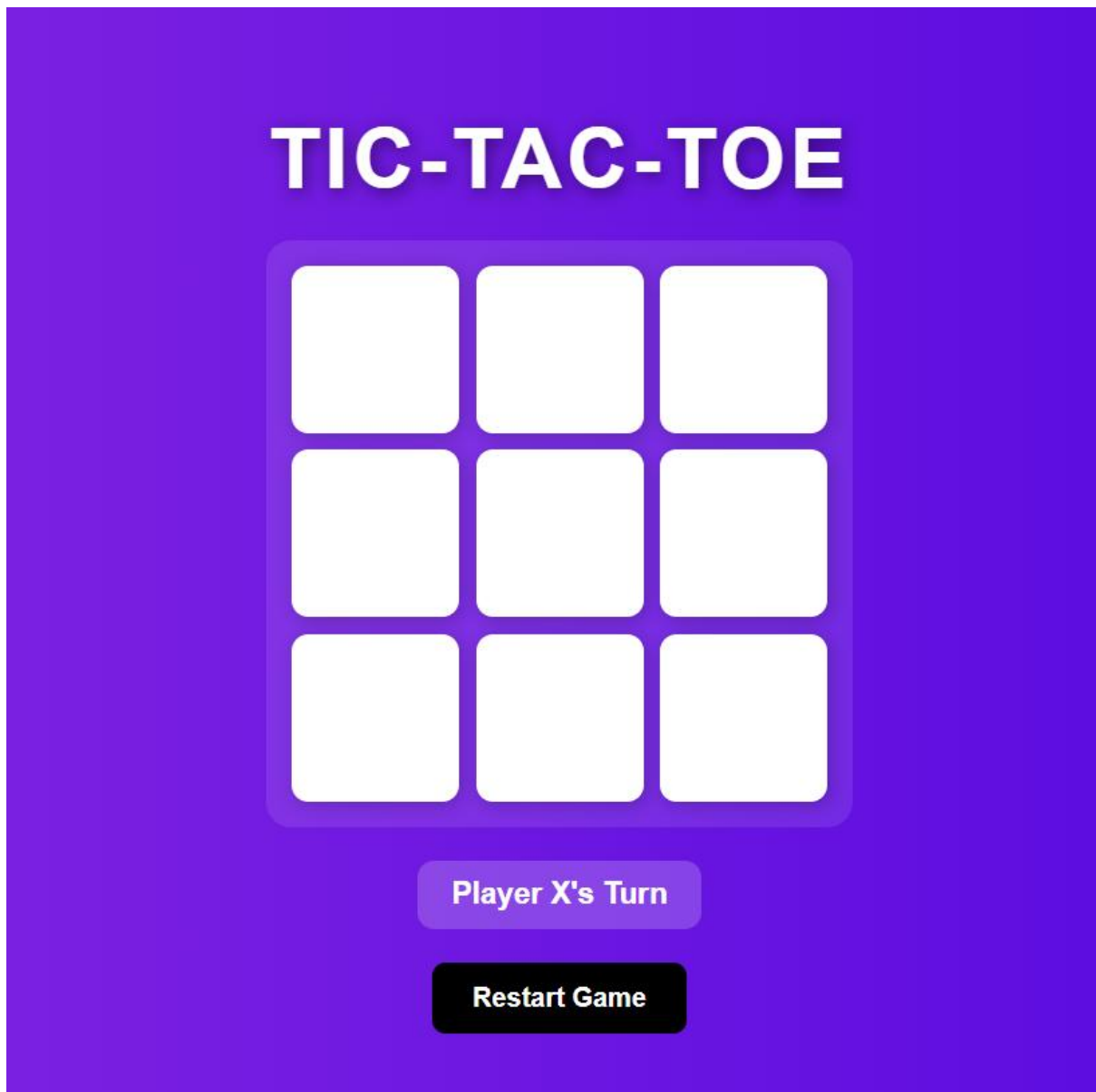
```

function restartGame() {
    board = ["", "", "", "", "", "", "", "", "", ""];
    cells.forEach(cell => cell.textContent = "");
}

```

```
gameActive = true;  
currentPlayer = "X";  
statusText.textContent = "Player X's Turn";  
}
```

🖼️ Screenshots:



TIC-TAC-TOE



Player X's Turn

Restart Game

TIC-TAC-TOE



Player O's Turn

Restart Game

❏ Conclusion:

- This project demonstrates the ability to build a fully functional, interactive browser-based game using core web technologies. It offers an engaging user experience with real-time feedback and polished design, making it suitable for learning and demonstrating front-end development skills.